

**NATIONAL PRACTICAL EXAMINATION FOR
TECHNICAL SECONDARY SCHOOLS (TSS)
SCHOOL YEAR 2024-2025**

SUMMATIVE/INTEGRATED ASSESSMENT

SECTOR: ICT AND MULTIMEDIA

TRADE: SOFTWARE DEVELOPMENT

**Integrated Assessment Project: Develop a web application using
Node JS, React JS and MySQL/MongoDB**

COMPETENCES ASSESSED

S/No	Code	Competence names	RQF Level
1	SWDPR301	Analyze Project Requirements	Level 3
2	SWDUX301	Design UI/UX	Level 3
3	SWDJF301	Apply JavaScript	Level 3
4	SWDDA401	Apply Data Structure and Algorithm Fundamentals Using JavaScript	Level 4
5	SWDBS401	Design Backend System	Level 4
6	SWDDD401	Develop Database	Level 4
7	SWDBD401	Develop a Backend Application using Node Js	Level 4
8	SWDFA501	Develop Frontend Application using React.JS	Level 5
9	GENQA501	Apply Quality Assurance	Level 5
10	SWDND501	Develop NoSQL Database	Level 5

1. INTEGRATED SITUATION

Scenario

SmartPark is company located in Rubavu District, western province of Rwanda. It provides multiple car-related services. It is struggling with inefficiencies due to its manual, paper based system for managing parking space sales. The human resource records employee details characterized by employee number, First name, Last name, address, position, Telephone, Gender, hired date and his or her department details characterized by code and department name. Based on department data human resource manually make payroll characterized by gross salary, total deduction, net salary and month of payment. This process is slow, prone to errors and makes it difficult to track employee's payment and generate monthly report efficiently. To address these challenges, the SmartPark needs a web based application that handles the Payroll making process. The system should allow the human resource to record employee details digitally and automatically generate needed reports.

Task: Employee Payroll Management System (EPMS)

As a full stack developer, you are given 7 hours to develop that web based application by:

1. Using attributes provided below, design an Entity Relationship Diagram (ERD) that represents the relationship between their entities.
 - Identify appropriate **primary keys** and **foreign keys** based on the relationships among the entities.
 - Entities and attributes are:
 1. **Employee** (employeeNumber, FirstName, LastName, Position, Address, Position, Telephone, Gender, hiredDate)
 2. **Department** (DepartementCode, DepartementName, GrossSalary)
 3. **Salary** (GlossSalary, TotalDeduction, NetSalary, month)
 - ERD should be drawn before using the computer.
 - ERD should be drawn on plain paper using pencils.
 - ERD should indicate cardinalities, relationships with correct symbols
2. Creating database called **EPMS** with Employee, Department and Salary tables as designed in ERD.

- Use this information for departments and Payroll marking

DepartementCode	DepartementCode	GrossSalary	Total deduction
CW	Carwash	300,000 rwf	20,000 rwf
ST	Stock	200,000 rwf	5,000 rwf
MC	Mechanic	450,000 rwf	40,000 rwf
ADMS	Administation Staff	600,000 rwf	70,000 rwf

3. Saving your work in your real names in a folder called **(FirstName_LastName_National_Practical_Exam_2025)**.
4. Preparing React.js Front-end application development environment by installing required modules and dependencies
5. Preparing JavaScript runtime environment for Node.js
6. Creating react.js components with UI features that will enable user to input his data in the tables above and display required reports.
 - Menu bar/Pages of web application should include Employee, Department and Salary, Reports and Logout options.
 - The web application should be responsive
7. Creating react.js components with UI features that will enable user to input his data in the tables above and display required reports.
8. Developing backend and frontend of parking sales management system.
 - Name backend project folder as **backend-project** and frontend project folder as **frontend-project**.
 - Use Tailwind CSS to implement UI design.
 - Your backend application should communicate to MySQL database/MongoDB to perform CRUD operations of using Node.js runtime environment and express.js framework.

Note: 1.Insert operation should be used on all four (4) forms (Employee, Department and Salary)

2. Delete, update, retrieve operations should only be used on Salary form
9. Creating react.js components with UI features that will enable user to input his data in the tables above and display required reports.

- 10.** Creating a session based login user account having username and password.
- 11.** Integrating your react.js application to backend application using Axios to input his data in the tables above and display required reports.
- 12.** Generating monthly Employ payroll composed by FirstName, LastName, Position, Departement and Net Salary.
- 13.** Removing permanently your project with its related configurations after being marked.
 - Ask permission to the assessor before removing your project.

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Note to the assessor:

- An assessor may promptly intervene to prevent the unnecessary use of internet and any other injury/incident that may occur during assessment
- The assessor should allow lunch time (one hour) break for candidate
- The assessor should ensure that each candidate is given time to exhibit his/her project by answering the following question:
 1. Describe your product in brief.
 2. Which difficulties have you met along the implementation of the project
 3. How did you overcome the difficulties
- The assessor should make sure that the distance between candidates prevents cheating
- The assessor should collect all ERD drawn by the candidates together with the exported project as evidences.

2. CONSUMABLE MATERIALS, TOOLS AND EQUIPMENT

2.1. Consumable materials

2.1. A. Reusable materials

SM	Item	Technical specifications	Unit	Quantity	Unit price
1	N/A				
2					

2.1. B. Non Reusable materials

SN	Item	Technical specifications	Unit	Quantity	Unit price
1	Internet	Minimum downloading speed of 25Mbps	mbps	1GB/2 Candidates	920
2	Flash disk	16 GB	Piece	1/20	475

2.2. Tools

2.2. A. First Choice (ideal) Tools

S/N	Item	Technical specifications	Measuring Unit	Quantity
1	VS Code	Latest Version	NA	1
2	Node	Stable Version	NA	1
3	Postman	Latest version	NA	1

2.3. B. Alternative Tools

SN	Item	Technical specifications	Measuring Unit	Quantity	Similar tools
1	Sublime text editor	Latest Version	NA	1	Any Text editor
2	Node	Stable version	NA	1	
3	Thunder client	Extension in vs code	NA	1	

2.3. Equipment

2.3. A. First Choice (ideal) equipment

SN	Item	Technical specifications	Measuring unit	quantity
1	Laptop or Desktop Computer	i5 and above, RAM: 4GB, HDD :500GB		1
2	Generator	Automatic		1

2.3.B. Alternatives equipment

SN	Item	Technical specifications	Measuring unit	Quantity	Similar Equipment
1	Laptop or Desktop Computer	Intel, I3; RAM: 4GB; HDD :500GB		1	
2	Generator	Manual		1	
3	UPS	8 hours of saving		1	

3. ASSESSMENT CHECKLIST

Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
1.Preliminary activities performance (15%)						
	1.1.ERD is properly designed based on system requirements	Department entity is drawn			1	
		Employee entity is drawn			1	
		Salary entity is drawn			1	
		Entity Symbol is used			1	
		Relationship symbol is used			1	
		Link Symbol is used			1	
		Primary Key rule respected			1	
		Foreign Key rule is respected			1	
		Cardinalities is indicated			1	
		Department and Employee relationship is indicated			1	
		Employee and Salary relationship is indicated			1	
		Department and Salary relationship is indicated			1	
		User Primary key is indicated			1	
		Department Primary key is indicated			1	
		Employee Primary key is indicated			1	
		Salary Primary key is indicated			1	
		Department Foreign key is indicated in Employee			1	
		Department Foreign key is indicated in Salary			1	
		Employee Foreign key is indicated in Salary			1	
Sub Total					.../19	
Percentage (15%)						
2.Process and fulfillment of the task (50%)						
	2.1Development environment is properly arranged based on coding architecture methodology	Project folder is named as FirstName_LastName_National_Practical_Exam_2025			1	
		Node.js project is created			2	
		Express.js is installed			1	
		Cors is installed			1	

Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
		Nodemon is installed			1	
		MySQL/MongoDB is installed in Node.js			1	
	2.2.React.JS environment is properly prepared based on application requirements.	React project is created			2	
		React-router-dom is installed			1	
		Axios is installed			1	
	2.3.Data definition language commands are effectively applied based on database schema	EPMS database is created			1	
		Users table is created			1	
		Department table is created			1	
		Salary table is created			1	
		Employee table is created			1	
		Primary keys are applied in user table			1	
		Primary key is applied in Department table			1	
		Primary keys is applied in Employee table			1	
		Primary keys is applied in Salary table			1	
		Department foreign key is applied in Employee table			1	
		Employee foreign key is applied in Salary table			1	
		Department foreign key is applied in Salary table			1	
	2.4. React basics are correctly applied based on application requirements.	Function is declared			2	
		Return method is included			1	
		Function is exported			1	
		Component is mounted to the DOM			2	
		JSX is used inside return method			1	
		JSX is well structured			1	
		Login form is created			1	
		Department registration form is created			1	
		Employee form is created			1	
		Salary form is created			1	
		Routes are configured			2	

Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
	2.5.UI navigation is correctly applied based on application requirements	Routing is performed			1	
		Links are created			2	
		Navigation between components is enabled			2	
	2.6.Tailwind utility classes are correctly Applied based on UI Design.	Tailwind CSS is installed			1	
		Tailwind CSS is configured			1	
		Tailwind is imported in the project			1	
		Class elements are used in JSX			2	
		Flexbox is applied			1	
	2.7.Responsive design principles are properly applied according to the required design	Flexible grid layouts are used			1	
		Media breakpoints are removed			1	
		Readability is Considered			1	
		Interactive elements are applied			1	
	2.8.Tailwind styles for a unique look are accurately customized based on tailwind configuration	Attractive colors are used			1	
		Text formatting is performed			1	
		UI components are attractive			1	
		Hover state is used			1	
		Focus state is used			1	
	2.9.Server and database connections are properly established according to development environment	Server JS file is created			2	
		Express.js package is imported			1	
		Cors package is imported			1	
		Port number is identified			1	
		Express object is created			1	
		Listen method is applied			1	
		MySQL/MongoDB package is used			2	
		Create connection method is used			1	
		Host parameter is passed			1	
		Username parameter is passed			1	
		Password parameter is passed			1	

Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
		Database parameter is passed			1	
		Messages are displayed			1	
		2.10.RESTFUL APIs are effectively implemented based on backend functionalities	POST endpoint is created			2
	GET endpoint is created				2	
	PUT endpoint is created				2	
	DELETE endpoint is created				2	
	Insert statement is used				2	
	Select statement is used				2	
	Update statement is used				2	
	Delete statement is used				2	
	2.11.API integration is properly implemented based on user requirements	API requests folder is created			2	
		Users API JS file is created			1	
		Spare_Parts API file is created			1	
		Employee API file is created			1	
		Salary API file is created			1	
		Axios is imported			1	
		API client is set			1	
		POST http method is used			2	
		Base URL parameter is passed			2	
	2.12.User Authentication, Authorization and Accountability (AAA) are carefully applied based on NPM Universal Access Control (UAC)	Form data is validated			2	
		Password input is encrypted			1	
		Duplicate data entry is eliminated			2	
		Password recovery is implemented			2	
Sub Total					/ 107	
Percentage (50%)						
3.Product presentation/Exhibition and Quality Assessment (30%)						
		The product name is mentioned			1	

Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
	3.1.Project is effectively presented	The key steps of the process are stated			3	
		The use/function/application/importance of the product is stated			2	
		Challenges/difficulties met are stated			1	
		Ways to overcome the challenges are stated			1	
		Candidate voice is audible			1	
		Body language is used			1	
		3.2.Usability is correctly tested according to expected results	Input validation prevents invalid data			2
	Form submission saves data				2	
	Data retrieval displays records				2	
	Update functionality works				2	
	Delete functionality works				2	
	3.3.UI navigation is correctly applied based on application requirements	Navigation menus are interactive			1	
		Buttons are interactive			1	
		Fonts style is consistent			1	
		Intuitive colors are used			1	
		UI colors are consistent			1	
		Monthly Employ payroll report is generated			3	
		Generated report is relevant			3	
Sub Total					.../31	
Percentage (30%)						
4.Closing activities (5%)						
	4.1.Project configurations and its dependencies are properly removed	Project folder is permanently removed			1	
		Database is deleted			1	
		Global dependencies are removed			1	
		Text editor is not uninstalled			1	
		XAMPP is not uninstalled			1	
		Paper is free of excessive erasures			1	

Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
	4.2. Best practices to keep the environmental clean are properly applied according to environmental laws, standards and regulations.	Waste materials are disposed off			1	
		Unnecessary creases or tears are avoided on papers used.			1	
	4.3. Time is properly managed	Time is respected			2	
Sub Total					.../10	
Percentage (5%)						
Total Marks					100	
Percentage						

4. PRE-DETERMINED CONDITIONS

General Environmental factors	Specific Equipment	Specific Tools
(1) Adequate Workshops and Laboratory space	-One desktop or laptop computers/student -Standby desktop or laptop computers for interventions in case another fails	Web browser, MySQL/MongoDB ,VS code/Sublime text editor
(2) Power source (Electricity + Generator)	Generator and UPS for those who use desktop computers	Automatic Generator and UPS of 1 hour power saving
(3) Availability of reliable internet and technological resources	Wired or Wireless internet	Cables or Modem
(4) Safety measures and protocol in place	The workspace should be designed to prevent liquids from coming into contact with electronic equipment, ensuring safety and longevity.	NA
(5) Office facilities (Computers, Printers, photocopying machines, Stapling machines)	Computers, Printers, photocopying machines, Stapling machines	Computer
(6) External storage device	External hard disk driver	NA